

LITAO LIU

Microelectronic Science and Engineering | Academic Curriculum Vitae

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github.com/Velliot-CD | velliott-cd.github.io

Microelectronics undergraduate with experience in processor RTL, analog circuit design, device simulation, embedded systems and applied AI projects. Current final-year research focuses on RISC-V instruction support for edge INT8 inference.

Education

University of Electronic Science and Technology of China

BEng in Microelectronic Science and Engineering | Glasgow College

Aug 2023 - Jun 2027 expected

GPA 3.55/4.00 | Weighted Average 83.38/100

IC and devices: Electronic Devices (99/100); Design of Integrated Circuits (95/100); Microelectronic Packaging Technology (90/100); Comprehensive Experiment of Modern Integration Technology (90/100); Circuit Analysis and Design (89/100); Semiconductor Physics (87/100).

Computing and systems: Introductory Programming; Microelectronic Systems; Embedded Processors; Application and Design of Digital Logic; Power Electronics; Probability Theory and Mathematical Statistics; Linear Algebra and Space Analytic Geometry.

Research Experience

RISC-V Fused Dot Product Accumulation for Edge INT8 Inference

Individual undergraduate final-year project | Vivado | Ongoing

Jul 2026 - Present

- Implemented operand collection and pipeline control for a fused four-lane INT8 dot-product-accumulate instruction in a five-stage RV32I core with two register-file read ports.
- Validated one MobileNet V1W convolution layer on ten inputs: each of three architectures matched 92,160 INT32 and 92,160 INT8 outputs in RTL and selected mapped functional simulations.
- Reduced mean layer cycles by 10.259% versus a DOT4-plus-ADD baseline at 50 MHz with 32 KiB RAM; completed out-of-context FPGA timing closure. Evaluation is single-layer simulation, with no board deployment.

Large Language Models for Supercomputing Infrastructure Operations

Third author of a six-author conference study

2025

- Primarily conducted the literature review and contributed to a limited portion of model setup for a study of retrieval-augmented generation (RAG) in infrastructure operations and maintenance.
- The study combined semantic retrieval and Cross-Encoder reranking over domain operating procedures, evaluating RAG and non-RAG systems with a four-level question bank.
- The team reported mean precision, recall, F1 and logical consistency of 0.92, 0.89, 0.90 and 0.88, respectively.

Publication: C. Sun, L. Ren, L. Liu, B. Sun, Y. Li, and Z. Liu, "Find the Limitation of LLMs in Supercomputing Center Infrastructure Operation and Maintenance," 2025 Global Reliability and Prognostics and Health Management Conference (PHM-Xi'an), 2025. doi:10.1109/PHM-Xian66756.2025.11427414

Industry Experience

Chengdu Kelichuang Intelligent Technology Co Ltd

Embedded Software Intern | R&D Department | Chengdu, China

Jun - Aug 2025

- Supported STM32 radar-module driver development and debugging of control and communication interfaces, including hardware-software integration and interface testing.
- Assisted with fault diagnosis and investigation of communication-latency issues, checking the relationship between software commands and hardware responses.

Analog Circuit and Semiconductor Projects

Two Stage CMOS Operational Amplifier with Extracted Layout Simulation

Individual coursework | CSMC 0.50 μm st02 | Cadence Virtuoso, Spectre and ADE

Mar - May 2026

- Completed a design sequence from NMOS characterization and a current-source-loaded amplifier to a Miller-compensated two-stage CMOS operational amplifier, including transistor sizing and bias analysis.
- Implemented matched, symmetric layout and obtained an LVS-matched extracted view for parasitic simulation. Spacing violations remained in the coursework DRC results.
- At the typical process corner with a 3 pF load, extracted-view simulation yielded 82.93 dB open-loop gain, 22.48 MHz unity-gain bandwidth, 52.4° phase margin and 1.195 mW power.
- Compared schematic and extracted-view results across tt, ff, ss, sf and fs process corners. Extracted-view gain ranged from 82.37 to 83.13 dB, unity-gain bandwidth from 20.37 to 22.48 MHz and power from 1.146 to 1.249 mW.

Supply Rejection Study of a Two Stage CMOS Operational Amplifier

Individual coursework extension | SMIC18MMRF | Nominal pre-layout simulation

2026

- Designed an NMOS-input, PMOS-current-mirror two-stage amplifier with a series-resistor Miller compensation network in Cadence Virtuoso and Spectre.
- Checked DC operating points and saturation of the principal gain and current-source transistors; obtained 85.96 dB low-frequency open-loop gain in nominal schematic simulation.
- Analyzed supply coupling near the 2.413 kHz open-loop -3 dB bandwidth point; the supply-to-output small-signal transfer was -79.54 dB under the coursework measurement convention.

Exploratory Device Simulation for Graphene FET Biosensing

Completed team project | Silvaco DeckBuild, TonyPlot and TonyPlot3D

Dec 2023 - Dec 2024

- Reviewed graphene field-effect transistor (GFET) biosensing mechanisms and contributed to a 13-page team research report.
- Built exploratory two- and three-dimensional device simulations, defining meshes, material regions and source, drain and gate electrodes.
- Swept gate and drain bias to examine output characteristics, transfer characteristics and electric-field distributions, while considering model limitations and thermal effects.

Power Electronics Laboratory Experience

Power Conversion and Control Laboratory Series

Individual coursework | LTspice, TL494, XL6009E1 and LM1117-3.3

Apr - May 2026

- Built and characterized a TL494 PWM generator; a 0.5-3.5 V control input varied the two outputs across approximately 9%-96% duty cycle.
- Tested rectifier and filter circuits; measured ripple decreasing from 1.40 to 0.74 Vpp as smoothing capacitance increased from 2.2 to 6.9 mF.
- Bench-tested an XL6009E1 boost module with a 5 V input at 10 V and 20 V outputs, measuring approximately 70 and 72 mVpp ripple, respectively; also characterized LM1117-3.3 voltage regulation under load.
- Simulated square-wave, bipolar-PWM and unipolar-PWM full-bridge inverters in LTspice, using Fourier analysis to compare output spectra and switching harmonics for a 60 Hz reference.

Digital Design and Embedded Systems Projects

CMOS Logic Cells and 28 Transistor Full Adder

Individual VLSI coursework | Electric 9.07 and IRSIM

Sep 2026

- Implemented supplied CMOS logic-cell and 28-transistor full-adder circuits, including schematics, transistor sizing, shared-diffusion layout and M1/M2 routing (drawn gate length 0.6 μm).
- Completed DRC, well/antenna ERC and schematic-layout NCC for the full adder with zero errors or warnings; physical layout bounds were 206 x 84 lambda (1,557.36 μm^2).
- Verified all eight full-adder input combinations through 16 recorded output pairs in extracted-layout IRSIM RC simulation. OR2 checks covered 13 samples and all 12 directed state transitions; delay, power and noise-margin characterization remain incomplete.

PT100 Temperature Measurement System Digital Backend

Digital RTL design and verification | Two-person digital subgroup in a six-member team

Apr - Jun 2026

- Co-developed a VHDL backend in Xilinx Vivado across five design units: Digital_Top, Out_of_Range_Detector, LED_Controller, Data_Packer and Serializer.
- Implemented range thresholds of 496 and 3599 ADC counts with 12-count hysteresis, a 1 Hz warning indicator, and packing of a 12-bit measurement plus 4-bit device ID into a 16-bit LSB-first serial frame.
- Contributed module- and top-level testbenches using five representative ADC codes and boundary cases; checked packed words, bit order, 16-cycle serialization, reset, handshaking and registered range flags.
- Contributed to team system validation; the report's five simulated temperature points had absolute errors of 0.034-0.084 $^{\circ}\text{C}$, within the 0.5 $^{\circ}\text{C}$ requirement.
- The project's Vivado synthesis and implementation reports recorded WNS > 0 and TNS = 0 under a 1 kHz design constraint.

CPU Design in Vivado

Completed team coursework | HDL design and simulation

2026

- Contributed to CPU HDL design, project integration and testbench simulation in Vivado, producing an integrated project and a repeatable simulation workflow.

Line Following Robot with a 24 GHz Radar Module

Hardware subsystem lead | Five-member Team Design Project

Mar - Jun 2026

- Designed and assembled a custom PCB around the STM32G431RBT6 using EasyEDA, integrating an eight-channel infrared sensor array, LoRa communication, dual-motor drivers and an HLK-LD2450 radar module.
- Separated motor and signal power supplies to reduce interference, and supported soldering, assembly and hardware-software integration across the vehicle.
- Investigated severe left drift through motor-driver output checks, wheel alignment and left/right PWM adjustments, helping restore stable line-following behavior.
- The team completed the core vehicle functions with reported project spending of RMB 2,061 against a RMB 2,500 budget.

Robotics Competition Project

Synchronized Performance by Two 18 DOF Hexapod Robots

Motion programming and synchronization | Three-member team

May - Jul 2026

- Contributed motion programming and debugging for two 18-DOF Libot hexapods, coordinating gait, posture and formation sequences; the team calibrated offsets for all 36 bus servos.
- Helped align motion sequences with music for a 3 min 12 s performance using the Zide motion editor, including short impact frames and longer sustained movements.
- The system used a one-shot Bluetooth trigger followed by local-clock execution to limit reliance on continuous wireless commands during the performance.
- The team received Second Prize in both the National Finals and Sichuan Regional Competition of the 28th China Robotics and Artificial Intelligence Competition.

International Academic Programmes

UWE Bristol International Telerobotics Summer School

Team practicum | Bristol Robotics Laboratory | Bristol, UK

20 - 31 Jul 2026

- Worked on a telerobotic manipulator under the supervision of Giacinto Barresi, contributing Python debugging and hardware-software integration.
- Integrated Leap Motion and gamepad inputs and investigated transmission-latency and pose-feedback issues; the team completed an object-transfer demonstration.

Pneumonia CT Classification at NTU Business AI Lab

Artificial Intelligence Internship Programme | Team project | Singapore

5 - 16 Aug 2024

- Contributed CT-image preprocessing, training-workflow setup and experimental result visualization for a comparison of CNN, ResNet and Vision Transformer classifiers.
- Completed workshops, assignments and the twelve-day team project, which received a Distinction rating.

Mathematical Modeling Competition

Mathematical Contest in Modeling

Problem C | Model development | Three-member UESTC team

2025

- Contributed mathematical model formulation for Problem C in a three-person team; the submission received an Honorable Mention from COMAP.

Honours and Awards

2026 | 28th China Robotics and Artificial Intelligence Competition - Second Prize, National Finals.

2026 | 28th China Robotics and Artificial Intelligence Competition - Second Prize, Sichuan Regional Competition.

2025 | Mathematical Contest in Modeling, COMAP - Honorable Mention.

2025 | Glasgow College, UESTC - Third-Class Academic Scholarship.

Technical Skills

Programming: Python, C/C++ and MATLAB, used in coursework and engineering projects.

Digital design and architecture: VHDL, SystemVerilog, Xilinx Vivado, RTL development, processor datapath and control, testbenches, synthesis, XDC constraints, timing analysis, data packing and serial interfaces; Electric layout, NCC and IRSIM RC verification.

Analog IC design: Cadence Virtuoso, Spectre and ADE; transistor sizing, bias analysis, Miller compensation, layout, DRC review, LVS, parasitic extraction and process-corner simulation.

Device modeling: Silvaco DeckBuild, TonyPlot and TonyPlot3D; exploratory device modeling, bias sweeps and interpretation of simulated device characteristics.

Embedded hardware: STM32, EasyEDA, PCB design, soldering and assembly, LoRa, sensor and motor interfaces, hardware-software integration and fault diagnosis.

AI and data analysis: CT-image preprocessing and CNN, ResNet and Vision Transformer model comparison; literature review and limited model-setup experience in RAG research.

Circuit simulation and laboratory work: LTspice, Fourier analysis, PWM and power-converter testing; oscilloscope, digital multimeter and function generator.

Languages

Mandarin Chinese: Native. English: IELTS Academic 6.0 overall (Listening 6.0, Reading 5.5, Writing 6.5, Speaking 6.5), 26 Apr 2026.